P-C-535B
January 10, 1978
SUPERSEDING
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September 22, 1966

FEDERAL SPECIFICATION

CLEANING COMPOUND, PLATERS' ELECTROCLEANING, FOR STEEL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for a heavy-duty anodic electrocleaner.

1.2 Classification. The cleaning compound covered by this specification shall be of one type and one grade.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

RR-S-366 - Sieves, Test.
PPP-D-729 - Drum, Shipping and Storage, Steel, 55 Gallon.

Federal Standards:

FED-STD-123 - Marking for Shipment (Civil Agencies).
FED-STD-536 - Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 6850

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

3. REQUIREMENTS

3.1 Materials. The cleanup compound shall conform to the chemical composition specified in table I and shall be free from fatty acids, rosin, starch, soaps, lignite inert fillers, and excessive dust-control agents.

3.2 Dust. When tested as specified in 4.5.3, the dust shall settle within a period of 5 seconds and there shall be no color change of the litmus paper.

3.3 Particle size. The maximum particle size of the ingredient material shall be such as to pass through a No. 4 sieve conforming to RR-S-366.

3.4 Performance.

3.4.1 Tarnishing of steel. The minimum allowable current density for tarnishing of steel shall be 150 amperes per square foot.

3.4.2 Conductivity. The conductivity (of a 6-percent by weight solution) of the cleaning compound shall be at least one-half that of a solution of NaOH of the same concentration.

3.4.3 Foam blanket. The foam blanket of the cleaning compound shall have a minimum thickness of 1/4 inch and a maximum thickness of 1-1/2 inches.

3.5 Workmanship. The cleaning compound shall be manufactured in accordance with the best commercial practice to produce a uniform, free-flowing and high quality material that is stable and not subject to change with age in a sealed container.

TABLE I. Chemical composition

Component	Minimum (% by weight)	Maximum (% by weight)	Test (paragraph reference)
Organic detergent (100 percent active basis)	0.4	----	4.5.2
Phosphate as P_2O_5 anhydrous	5.0	----	4.5.1
Silicate as SiO_2	10.0	35.0	4.5.1
Impurities:			
Carbonates	----	2.0	4.5.1
Chlorides	----	1.0	4.5.1
Insolubles (in distilled water 100°C)	----	0.5	4.5.2.4
Moisture	----	2.0	4.5.1
Sulfates and sulfites calculated as SO_4	----	2.0	4.5.1
Caustic soda	Remainder		

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Quality conformance inspections. The examination and testing of the cleaning compound shall be classified as quality conformance inspections.

4.2.1 Lot. A lot shall consist of all compound manufactured as one batch and offered for delivery at one time.

4.3 Sampling.

4.3.1 Examination. A random sample of filled containers shall be selected for examination in accordance with level I of MIL-STD-105.

4.3.2 Tests. From each lot one original unopened container shall be selected at random and samples taken to perform the tests specified in 4.6.

4.4 Examination.

4.4.1 Visual. Sample units selected in accordance with 4.3.1 shall be examined for defects at the acceptable quality level (AQL) of 1.5 and inspection level I of MIL-STD-105.

4.4.1.1 Appearance of cleaning compound. The sample unit for this examination shall be one filled primary container.

Examine	Defect
Appearance of cleaning compound	Not uniform. Not free from lumps. Not free-flowing powder. Presence of foreign matter, grits, or abrasives.
Odor	Any strong or objectionable odor.

4.4.1.2 Containers. The sample unit for the examination of the end item containers for defects in appearance, closure, construction, workmanship, and markings shall be one filled primary container.

Examine	Defect
Appearance, construction and workmanship	Not in container specified. Cracked, crushed, or any defect affecting serviceability. Not clean, any evidence of dirt, rust, or foreign matter.
Closure	Any leakage or seepage of contents. Closure not as specified.
Markings	Omitted, illegible, incorrect, or not in accordance with contract requirements.

4.5 Testing of the end item.

4.5.1 Samples of the cleaning compound shall be selected in accordance with 4.3.2 and shall be tested in accordance with FED-STD-536 and 4.5.2, 4.5.3, and 4.5.4.

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Test	FED-STD-536, Method No.
Silicates	1301
Phosphates	1501
Carbonates	1401 or 1402
Chlorides 1/	501
Sulfates and sulfites 2/	1601
Moisture	202
Sieve tests 3/	2101

1/ Test to be performed on dry matter insoluble in alcohol.
2/ Test to be performed on dry matter insoluble in alcohol. The following change will be made in Method No. 1601: After neutralizing carefully with hydrochloric acid to the methyl purple end point, add an excess of bromine water, then add 5 milliliters excess of hydrochloric acid, etc.
3/ Section 2 of Method 2101 is applicable.

4.5.2 Determination of organic detergent. The Weatherburn Modified Epton method shall be used for determining the anionic surface active agents.

4.5.2.1 Reagents.

(a) 0.002M Cetyltrimethylammonium bromide (CTAB). The CTAB should be filtered after making up to 0.002M (see 6.5).
(b) 0.010 M $K_2Cr_2O_7$ (reagent grade-dried 1 hour at 105°C (221°F)).
(c) Indicator solution:

0.03 gram methylene blue chloride)
12.0 grams concentrated H_2SO_4) dilute to 1 liter with
50.0 grams Na_2SO_4 anhydrous) distilled water

(d) 0.10 N sodium thiosulfate solution
(e) Chloroform
(f) Concentrated hydrochloric acid - reagent grade
(g) Dilute sulfuric acid (1:1)
(h) Potassium iodide - reagent grade
(i) Sodium bicarbonate - reagent grade
(j) Freshly boiled distilled water
(k) Starch indicator solution

4.5.2.2 Standardization of CTAB. CTAB shall be standardized as follows:

(a) Pipette 50 milliliters (ml) of CTAB solution into a 150-ml beaker and add to this by pipetting 10 ml of 0.010 M $K_2Cr_2O_7$ solution.
(b) Add 2 ml of concentrated hydrocholoric acid, stir the mixture thoroughly, and allow to stand for 15 minutes for the CTAB-dichromate precipitate to coagulate.
(c) Cool, filter through the Whatman No. 40 or 42 paper and wash precipitate and paper well with distilled water at room temperature. The excess dichromate is titrated iodemetrically.
(d) Place 4.0 grams KI and 2.0 gm $NaHCO_3$ in a 500-ml iodine flask containing 75 ml water and dissolve.
(e) Add 5 to 6 ml concentrated HCl slowly.
(f) Add the CTAB-dichromate filtrate and rinse well with a few ml of freshly boiled distilled water and set aside in the dark for 5 to 6 minutes.
(g) Cool the solution and dilute to 300 to 400 ml with cold, boiled distilled water.
(h) Titrate with the 0.10 N thiosulfate and add 1 to 2 ml starch indicator solution at the greenish stage, and then finish the titration.

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The standardization should be run in duplicate.
Calculation:
Molarity of CTAB solution =

$$\frac{(ml_{K_2Cr_2O_7} \times M_{K_2Cr_2O_7} \times 0.294) - (ml_{Na_2S_2O_3} \times N_{Na_2S_2O_3} \times 0.049) \times (2000)}{ml_{CTAB} \times (294.2)}$$

4.5.2.3 Method. The organic detergent in the cleaner shall be determined as follows:

(a) Weight 100 grams of cleaner to the nearest 0.1 gm, dissolve in distilled water and make up to one liter in a volumetric flask.
(b) Pipette 5 and 10 ml aliquots of the cleaner solution into 250-ml glass-stoppered cylinders or Almquist flasks.
(c) Add 2 ml of 1:1 sulfuric acid to the flask or cylinder containing the 10-ml aliquot and 1 ml of acid to the other.
(d) Make up volume to 50 ml with distilled water.
(e) Add 25 ml of indicator solution and 15 ml of chloroform.
(f) Shake the mixture well and let stand for several minutes.
(g) From a 10-ml burrette, titrate the samples with the standardized CTAB, shaking the cylinder vigorously after each addition. The endpoint of the titration is taken at the point at which the color of the two layers is the same when viewed by reflected light. The flask is allowed to stand 1 minute before making the color comparison.

Calculation:
Titration blank = $T_{10} - 2T_5$, where T_{10}, T_5 = ml.

CTAB required for titration of 10- and 5-ml aliquots of sample.
Corrected ml CTAB = T_{10} + titration blank; T_5 + titration blank.

$$\text{Percent organic S} = \frac{\text{Corrected ml CTAB} \times \text{MCTAB} \times 0.0321 \times 100}{\text{Weight of sample in aliquot}}$$

An estimation of the amount of active alkylbenzene sulfonate (or alkyl sulfate) may be made by assuming a molecular weight of 348 for the detergent used in the cleaner.

(Molecular weight dodecylbenzene sodium = 348).

$$\text{Percent alkylbenzene sulfonate or alkylsulfate} = \text{Percent organic S} \times \frac{348}{32.1}$$

4.5.2.4 Determination of solubility. Accurately weigh a 6-gram sample of the cleaner into a 250-ml beaker. Add 100 ml of boiling distilled water, stir to dissolve the sample and continue to boil the solution for 10 minutes. Any water lost by evaporation shall be replaced. Filter the solution through a tared Gooch crucible with an asbestos mat. Wash the residue with 100 ml of distilled water at a temperature not less than 82°C (180°F). Dry the crucible at 105°C (221°F) for 2 hours, cool, and weigh. The matter insoluble in distilled water shall be calculated as percent by weight.

4.5.3 Dust forming. The inside walls of a 250-ml glass-stoppered graduate shall be rendered completely free of any grease or moisture. A 25-garm sample of the cleaner shall then be placed on the bottom of the clean tube. The tube shall then be stoppered, inverted, and immediately returned to its original position. After the suspended dust has been allowed to settle of 5 seconds, the stopper of the tube shall be removed and a moist piece of red litmus paper suspended in the tube without touching the walls so that the lower end of the litmus paper strip coincides with the 210-ml mark at the upper end of the graduate. After 60 seconds immersion, the litmus paper shall be removed and examined for any color change.

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4.5.4 Performance.

4.5.4.1 Tarnishing. A steel tank having the approximate dimensions, 1 by 1 by 1 foot shall be filled with 5 gallons of a 6-percent by weight solution of the cleaning compound, and maintained at a temperature of 83°C (180°F). A test specimen of steel drill rod, high carbon steel, approximately 3.9 inch diameter shall be immersed in the cleaning solution at the center of the tank to a depth of 5 inches. The specimen rod shall be polished to a bright finish by using successively fine abrasives, the last of which shall be not coarser than No. 400 grit. With the test specimen as the anode and the tank as the cathode, the specimen shall be cleaned, using an anode current density on a test specimen of 150 amperes per square foot for 10 minutes. The test specimen shall then be removed and visually inspected for tarnishing.

4.5.4.2 Conductivity. The electrical conductivity of a 6 percent by weight solution of the cleaning compound shall be measured, using a conductivity cell suitable for measuring highly conductive solutions. Boiled distilled water shall be used and the temperature closely controlled at 25°C (77°F). Resistance measurements shall be made using a suitable alternating current Wheatstone bridge. The following formula shall be used in computing conductivity (mhos):

$$L = \frac{1}{R}$$

Where: L = conductivity in mhos.
R = resistance in ohms.

The conductivity of a 6 percent by weight solution of reagent grade NaOH, weight to volume, using boiled distilled water, shall be obtained using the same conductivity cell. The conductivity obtained shall be used as a basis of comparison.

4.5.4.3 Foaming. The apparatus and conditions specified in 4.5.4.1 shall be used for foaming tests except that a steel panel, 4 by 4 inches, shall be used as the anode. Forty amperes per square foot of solution surface area of the tank shall be passed through the solution for 10 minutes. The foam thickness shall then be determined by averaging the foam thickness at the minimum and maximum points.

4.6 Packaging, packing, and marking. Preparation for delivery shall be inspected for conformance to section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Unless otherwise specified, the compound shall be packed in 55 gallon drums conforming to PPP-D-729. The type and size of containers and levels of packaging and packing shall be as specified by the procuring activity (see 6.2).

5.2 Interstate Commerce Commission Code of Federal Regulations shall govern in any conflict between the requirements specified herein and ICC regulations.

5.3 Marking.

5.3.1 Civil agencies. Shipping containers shall be marked in accordance with FED-STD-123.

5.3.2 Military activities. In addition to any special marking required by the contract or order, shipments to the Department of Defense shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The compound covered by this specification is intended primarily for the anodic cleaning of steel prior to electroplating and chemical treatments. It may be used for cathodically cleaning magnesium prior to chemical and electrochemical treatments.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Levels of packaging and packing (see 5.1).

6.3 Directions for use. The compound covered by this specification is generally used in concentrations from 8 to 12 ounces per gallon at a minimum solution temperature of 83°C (180°F).

6.4 Reagent No. T-5650 supplied by the Eastman Kodak Company has been found satisfactory.

MILITARY INTEREST:

Custodians:

Army - AV
Navy - SH
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Reviewers:

Army - MR
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